

Using SPARTAN and mpi4py
Workshop Week 3

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Getting Going on SPARTAN

- Karaage
 - Ball in your court
- ssh – secure shell (multiple sessions possible)
 - Putty.exe for Windows users (or other solutions)
- scp – secure copy
 - Move stuff (code) onto/off SPARTAN
- Basic Linux
 - ...enough to get you going in the assignment
- Text editor (if want to develop on SPARTAN)
 - vi
 - nano
 - ...

mpi4py

- /apps/examples/Python/MPI
 - Show and tell of typical examples
 - Note that SLURM scripts will need to be modified
 - Based on older modules

SPARTAN Software

- If you want to install ijson or numpy or ... then use a virtual environment
- For example:

```
module load GCC/11.3.0
```

```
module load OpenMPI/4.1.4
```

```
module load mpi4py/3.1.4
```

```
python -m venv myenv
```

```
source myenv/bin/activate (activate environment  
and update path)
```

```
pip install ijson (or numpy or...)
```

```
python myscript.py (script can include ijson etc)
```

```
deactivate
```

Assignment 1 - tips

- Small/medium are easy
 - Big can cause problems
 - Need to open/read big files without reading into memory
 - Out of memory exception
- Many solutions
 - Test on small/medium first
 - Then say why you chose the solution you did in your report for the big files
- Don't benchmark the solution on the big files

Assignment 1 - tips

- For 2:4 you can get the same nodes
 - You can force the SLURM to use separate nodes

```
#SBATCH --nodes=2
```

```
#SBATCH --ntasks-per-node=4
```

```
#SBATCH --ntasks=8
```